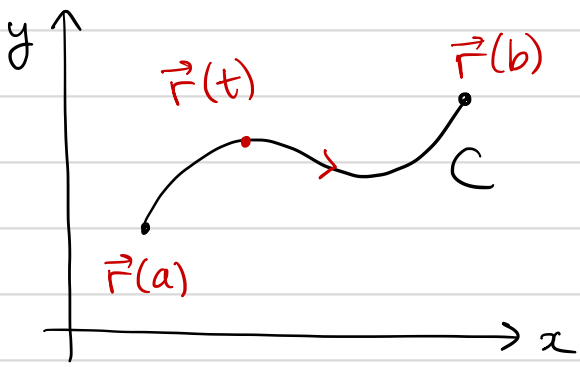
Lecture 12

- Parametrizations of curves
- Line integrals (16.2)

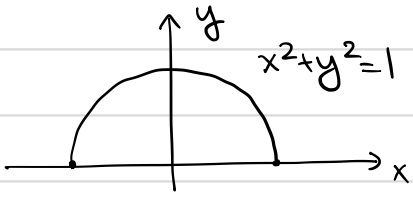


<u>Def</u>: $\vec{r}:[a,b]\longrightarrow\mathbb{R}^2$ is a <u>parametrization</u> of C if

$$\vec{r}(t)=(x(t),y(t))$$

is in C for all $t\in[a,b]$, and traverses C <u>once</u> as t increases from a to b.

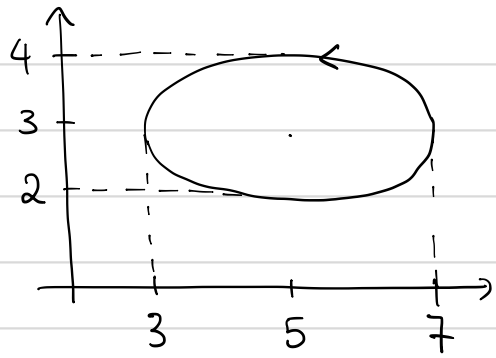
<u>Examples</u>: 1) Half circle



$$\left.\begin{array}{l} x=\cos t \\ y=\sin t \end{array}\right\} \quad \begin{array}{l}\vec{r}(t)=\langle\cos t,\ \sin t\rangle \\ (0\leq t\leq\pi)\end{array}$$

2) Ellipse

$$\frac{1}{4}(x-5)^2+(y-3)^2=1$$



$$\left\{\begin{array}{l} x=5+2\cos t \\ y=3+\sin t \end{array}\right.$$
$$0\leq t\leq 2\pi$$

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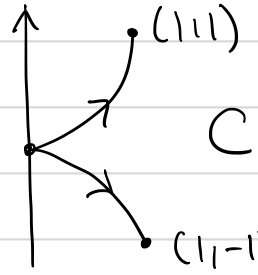
3) Cusp.

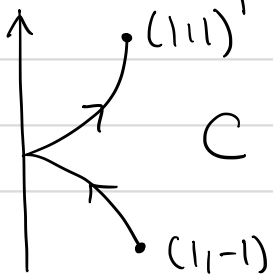
$y^2=x^3$



$C=C_1\cup(-C_2)$

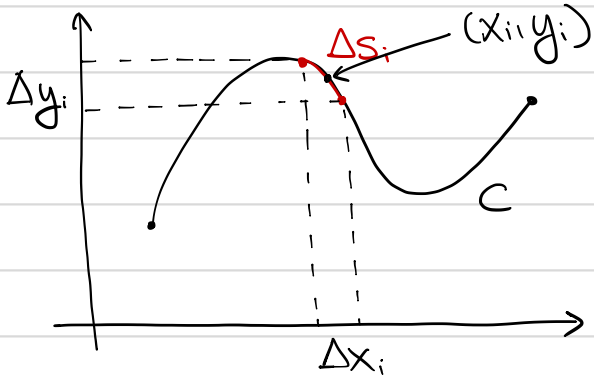
$$C_1:\left\{\begin{array}{l} x=x \\ y=+\sqrt{x^3} \end{array}\right. \quad (0\leq x\leq 1) \qquad C_2:\left\{\begin{array}{l} x=x \\ y=-\sqrt{x^3} \end{array}\right. \quad (0\leq x\leq 1)$$

3') A better parametrization:



$$\left\{\begin{array}{l} x=t^2 \\ y=t^3 \end{array}\right. \quad (-1\leq t\leq 1)$$

Line Integrals (2D)



Parametrization:

$$\vec{r}(t) = (x(t), y(t)), \quad a \leq t \leq b.$$

$$\text{length}(C) = \lim_{\Delta s_i \to 0} \sum_i \Delta s_i$$

$$= \int_a^b \underbrace{\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt}_{ds \text{ "arc length element"}}$$

Why:

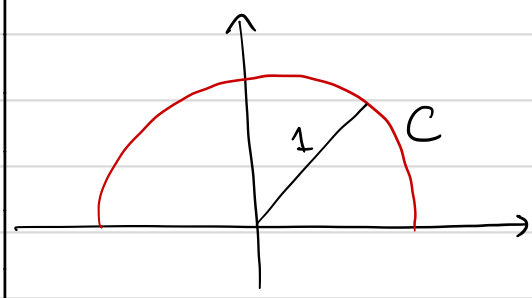
$$\Delta s_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$$

$$= \sqrt{\left(\frac{\Delta x_i}{\Delta t}\right)^2 + \left(\frac{\Delta y_i}{\Delta t}\right)^2}\ \Delta t$$

Definition:

$$\boxed{\begin{aligned} \int_C f(x,y)\, ds &= \lim_{\Delta s_i \to 0} \sum_i f(x_i, y_i)\, \Delta s_i \\ &= \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \\ &= \int_a^b f(\vec{r}(t))\, |\vec{r}\,'(t)|\, dt \end{aligned}}$$

Example: $f(x,y) = 2 + x^2 y$



$x = \cos(t)$

$y = \sin(t)$

$0 \leq t \leq \pi$

$$\int_C f(x,y)\, ds =$$

$$= \int_0^\pi (2 + \cos^2 t \sin t) \underbrace{\sqrt{(\sin t)^2 + (-\cos t)^2}}_{1}\, dt$$

$$= 2t - \frac{\cos^3 t}{3}\Big|_0^\pi = 2\pi + \frac{2}{3}$$

Applications of $\int_C f\,ds$:

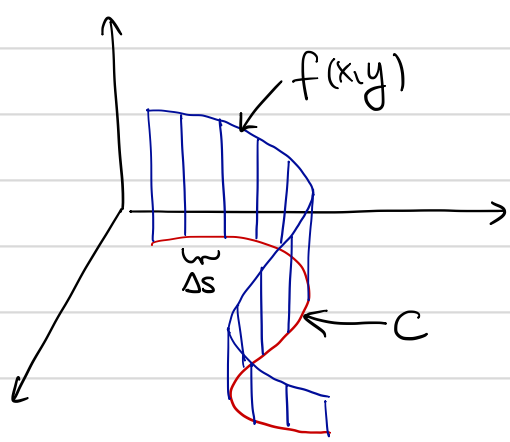
1) $\int_C ds$ = length of $C =: L$

2) Average value of f on C $= \frac{1}{L}\int_C f\,ds$

3) $\rho(x,y)$ = density of C (mass / unit length) at (x,y)

$$M = \int_C \rho\,ds$$

$$\bar{x} = \frac{1}{M}\int_C x\rho\,ds$$

$$I_y = \frac{1}{M}\int_C (x^2+z^2)\rho\,ds$$

4) Area of "fence":